



CRATE IQ: 'Credits-First' Typography Design Specification

The CRATE IQ user interface is designed for high-speed metadata consumption in variable lighting environments. The "Credits-First" layout prioritizes personnel and production data using a high-contrast, monochrome typographic system inspired by **Braun (Dieter Rams)** and **Teenage Engineering**.

1. Visual Specification (400x240px)

The following UI mockup demonstrates the finalized layout on the 2.7-inch Sharp Memory LCD.

Representative 'Credits-First' UI:



2. Typographic System: 'Crate-Sans'

To achieve the requested Design DNA, we have specified a custom-tuned version of a geometric Swiss sans-serif (rational, modular, and high-legibility).

Font Weights & Stroke Rules

- Minimum Stroke: 2 Pixels.** To avoid "wispy" or unreadable text on the reflective non-backlit 400x240 screen, all character strokes must be at least 2px thick [Play](#).
- Aliasing:** No anti-aliasing. The system uses a **1-bit (Black/White)** rendering engine for maximum sharpness and lowest power consumption.

Hierarchical Grid

Element	Font Size (px)	Weight	Alignment
Album / Artist Header	28px	Bold	Left-Aligned
Personnel Names	18px	Bold	Left-Aligned
Production Roles	14px	Regular	Left-Aligned
Contextual Footer	12px	Regular	Center / Split

3. Layout & Interaction Rules

The "Credits-First" Flow

- Header Lock:** The Album/Artist header is locked to the top 40px of the screen. A 2px divider line separates it from the body.
- Scrollable Personnel:** The body of the screen is a scrollable vertical list of personnel. Names are bolded to act as visual anchors, while roles are smaller and regular weight to reduce clutter.
- Digger-Logic Padding:** 8px horizontal padding from screen edges ensures text isn't obscured by the physical enclosure bezel.
- Instant-Facts Footer:** A permanent 20px footer bar displays the Record Label and Release Year, providing immediate context for "blind-buy" judging.

4. Implementation Guidelines for ODMs

- Display Buffer:** The microcontroller (e.g., NXP i.MX 8M Nano) must buffer the full 12KB image (400x240@1-bit) before writing to the Sharp Memory LCD [Panoxdisplay](#).
- Font Conversion:** Use a high-fidelity TrueType-to-Bitmap conversion process (e.g., Peekay123) to preserve character integrity at specific pixel heights [Particle](#).
- Refresh Rate:** Optimize for static viewing. The Sharp Memory LCD supports 1Hz refresh for metadata review, extending the CRATE IQ battery life to 24+ hours in "read-only" mode [Adafruit](#).

Next Steps

- Firmware UI Build:** Implement the 'Crate-Sans' bitmap font library in the device's firmware.
- User Legibility Test:** Conduct real-world "crate-side" legibility tests to confirm the 14px roles remain readable in low-light environments.
- Haptic Paging:** Integrate the side-dial scroll haptics with the vertical UI list to provide 1:1 tactile feedback for each personnel credit scrolled.

